

A Printable Asymmetric MXene Hybrid Lithium-Ion Cell: Materials Selection, Constraints, and a Conservative Design

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Abstract

This paper selects, from a family of printed MXene hybrid recipes, the subset of materials that is consistent with established electrochemistry and then writes an explicit cell around that subset. The recommended architecture is asymmetric. The negative electrode is a printed $\text{Ti}_3\text{C}_2\text{T}_x/\text{Nb}_2\text{CT}_x$ MXene composite with a nitrogen-doped graphene-nanoribbon spacer. The positive electrode is a distinct $\text{V}_2\text{O}_5/\text{TiO}_2@\text{Ti}_3\text{C}_2\text{T}_x$ hybrid. The separator is a cross-linked PVDF-PEO membrane containing LiTFSI, approximately 10 wt% surface-modified LLZO, and at most 5 wt% MXene—well below electronic percolation. Lithium inventory is provided by prelithiation of the negative electrode or by a sacrificial lithium source during formation. The design is a lithium-ion cell, not a lithium-free cell.

No new experimental cell is reported here. Capacities, voltages, and conductivities are taken from peer-reviewed half-cell and composite-electrolyte literature. A bill-of-materials estimate for a 150–180 μm printed stack places realistic cell-level energy density in the range of approximately 80–180 Wh kg^{-1} depending on matching, porosity, and irreversible loss—not 400–650 Wh kg^{-1} . Rate capability is expected to be good relative to thick polymer cells if electrodes remain thin; 10–15C at high energy density is not projected. The paper states the constraints that any subsequent prototype must satisfy and the measurements that would falsify the design.

Keywords: MXene; $\text{Ti}_3\text{C}_2\text{T}_x$; Nb_2CT_x ; V_2O_5 ; composite polymer electrolyte; LLZO; printed battery; lithium-ion

1 Purpose and scope

Several draft architectures combined $\text{Ti}_3\text{C}_2\text{T}_x$, Nb_2CT_x , $\text{V}_2\text{O}_5/\text{TiO}_2@\text{MXene}$, N-doped graphene nanoribbons, PVDF-PEO, LLZO, and aligned MXene into a single printed stack and assigned that stack cell-level energy densities of 400–650 Wh kg^{-1} , charge rates of 10–17C, and cycle lives from 10^3 to 3×10^5 . Those figures do not follow from the materials list. This paper keeps the materials that have experimental support, discards the combinations that violate charge neutrality, electronic insulation of the separator, or thermal stability of the polymer, and writes a cell that could be built and tested.

The paper is a design and literature analysis. It is not a claim of a demonstrated device.

2 Constraints that any such cell must satisfy

A rechargeable cell requires (i) a defined working ion, (ii) two electrodes with different operating potentials, (iii) a source of that ion sufficient for the first-cycle inventory and irreversible loss, and (iv) a separator that is ionically conductive and electronically insulating.

MXenes and V_2O_5 are lithium hosts in the literature cited below. Using them without a lithium source—while also using LLZO and, implicitly or explicitly, LiTFSI—is inconsistent. The cell described here is a lithium-ion cell. “Lithium-free” is not used.

Identical printed slurries on both current collectors do not produce a 2.8–3.2 V battery. $Ti_3C_2T_x$ and Nb_2CT_x operate at low-to-mid potential versus Li/Li^+ . Amorphous TiO_2 also operates near 1.5–1.8 V versus Li/Li^+ . V_2O_5 can operate as a mid-voltage positive electrode (~ 2 –4 V versus Li/Li^+). Those hosts must sit on opposite sides of the separator.

$Ti_3C_2T_x$ is electronically metallic (typically 10^3 – 10^4 S cm^{-1}). In polymer matrices, electronic percolation of MXene occurs at low loading. Composite-electrolyte studies that use MXene as a filler keep the loading near 1–5 wt% so that electronic conductivity remains negligible. Loadings of 35–38 wt% with intentional alignment are not separators; they are conductors. That constraint is treated as non-negotiable.

PEO melts near 65 °C and undergoes chain scission well below 400 °C. PVDF is more stable but dehydrofluorinates on the order of 300–360 °C. A 410 °C laser anneal of a PVDF–PEO–LiTFSI film is outside the process window of the electrolyte. Electrode drying and lamination must remain inside that window.

3 Recommended cell

3.1 Negative electrode

Composition (wt% of dry electrode solids):

- 55 wt% delaminated $Ti_3C_2T_x$ (preferentially –O-rich, fluorine-lean terminations);
- 25 wt% delaminated Nb_2CT_x ;
- 8 wt% nitrogen-doped graphene nanoribbons;
- 12 wt% PVDF binder (or a PVDF–PEO binder compatible with the electrolyte solvent).

Target loading for a first prototype: 2–4 mg cm^{-2} active mass, porosity 30–40%, thickness ~ 30 –50 μm after drying and calendaring.

$Ti_3C_2T_x$ is the standard MXene Li host. Reversible half-cell capacities for ordinary films are typically 100–250 mAh g^{-1} ; optimized or pillared films can be higher at low rate [Naguib et al.(2012), Lukatskaya et al.(2013), Cao et al.(2022)]. Theoretical capacity of bare Ti_3C_2 is about 320 mAh g^{-1} ; –F and –OH terminations reduce that value [Tang et al.(2012)]. Nb_2CT_x has a higher theoretical Li capacity than $Ti_3C_2T_x$ and has delivered approximately 170 mAh g^{-1} as-prepared and 300–450 mAh g^{-1} after Li pre-intercalation or –O enrichment [Naguib et al.(2013), Zhao et al.(2021)]. Mixing the two MXenes is intended to reduce restacking and add a second interlayer spacing,

not to create a positive electrode. Nitrogen-doped nanoribbons are a spacer and electronic additive; 8 wt% is a conventional level.

For design calculations, a reversible negative-electrode specific capacity of 180 mAh g⁻¹ (active mass) at low rate is used. First-cycle irreversible loss of 20–40% should be assumed until measured.

3.2 Positive electrode

Composition (wt% of dry electrode solids):

- 70 wt% V₂O₅/TiO₂@Ti₃C₂T_x hybrid (MXene fraction inside the hybrid kept low, on the order of 5–15 wt% of the hybrid, following [Myint et al.(2024)]);
- 10 wt% Ti₃C₂T_x or carbon additive if the hybrid itself is MXene-poor;
- 8 wt% nitrogen-doped graphene nanoribbons;
- 12 wt% PVDF binder.

Target loading: capacity-matched to the negative electrode after first-cycle loss, typically 3–6 mg cm⁻² hybrid, thickness ~40–60 μm.

[Myint et al.(2024)] reported ~460 mAh g⁻¹ at 0.1 A g⁻¹ and ~290 mAh g⁻¹ at 1 A g⁻¹ for a V₂O₅/TiO₂@Ti₃C₂ composite versus Li metal, with mixed capacitive and intercalation storage and substantial fade at high rate (62% retention over 500 cycles at 1 A g⁻¹). That result is a half-cell number. In a full cell the usable capacity will be lower. V₂O₅ can be used in a cathode window (~2–4 V versus Li/Li⁺) or driven into a conversion-like low-voltage window; only the former is useful opposite an MXene negative electrode.

For design calculations, a reversible positive-electrode specific capacity of 200 mAh g⁻¹ (hybrid mass) between approximately 2.0 and 3.8 V versus Li/Li⁺ is used until full-cell data exist. Average full-cell voltage after matching is estimated at 1.8–2.4 V, not 3 V.

Amorphous TiO₂ as a majority co-host on both electrodes is not used. TiO₂ is an anode material. It may remain as a minority phase inside the V₂O₅ hybrid, as in [Myint et al.(2024)].

3.3 Lithium inventory

Both printed electrodes start empty of lithium. The electrolyte salt is not a sufficient reservoir for a practical cell. One of the following must be chosen before assembly:

1. electrochemical or chemical prelithiation of the MXene negative electrode;
2. a sacrificial lithium-rich additive in the positive electrode that is consumed on the first charge;
3. a thin lithium or lithium-alloy interlayer used only for formation and then stripped, leaving a rocking-chair cell.

Option 1 is the cleanest for a printed solid cell. Without an explicit inventory the cell is not a battery.

3.4 Electrolyte / separator

Composition (wt% of dry membrane):

- 70–75 wt% cross-linked PVDF–PEO (approximately 60:40 by mass), with LiTFSI at an EO:Li ratio near 12–20;
- 10–15 wt% surface-modified cubic LLZO or Ta-doped LLZTO nanoparticles;
- 2–5 wt% $\text{Ti}_3\text{C}_2\text{T}_x$ (not aligned into a percolating sheet network through the thickness);
- optional ≤ 10 wt% plasticizer (e.g. succinonitrile or a low vapor-pressure carbonate) only if room-temperature conductivity is otherwise inadequate—this makes the membrane a quasi-solid electrolyte and must be stated as such.

Target thickness: 20–40 μm . Target room-temperature ionic conductivity: 10^{-4} S cm^{-1} class. Electronic conductivity must be measured by DC polarization and remain $\ll$ ionic conductivity (order 10^{-9} S cm^{-1} or lower).

PVDF–PEO–LLZO composites with lithium salt are a standard solid or quasi-solid electrolyte family, with reported room-temperature conductivities typically 10^{-4} to low 10^{-3} S cm^{-1} when the polymer is sufficiently amorphous and the ceramic is well dispersed [Zhao et al.(2021b), Li et al.(2024)]. Small MXene additions to PEO–LiTFSI retard crystallization and can raise conductivity at loadings of a few weight percent; at those loadings electronic conductivity remains negligible [Pan et al.(2019)]. That is the only MXene-in-electrolyte regime used here.

3.5 Current collectors

Commercial 8–12 μm Al foil on the positive electrode. Al or carbon-coated Al, or a thin printed carbon film, on the negative electrode. A dilute MXene dip-coat on the foil is acceptable as a contact layer. Both foils Al is acceptable for a prototype if corrosion and alloying at the negative side are checked after cycling.

3.6 Architecture that is not used

The following combinations from earlier drafts are excluded: symmetric electrodes; 35–38 wt% MXene in the separator; deletion of LLZO and of Li salt; 410 $^{\circ}\text{C}$ sintering of the polymer stack; cell-level energy-density claims above 250 Wh kg^{-1} from this bill of materials; cycle-life claims above what the V_2O_5 hybrid has shown in half-cells.

4 Manufacturing window

Inks may be slot-die or aerosol-jet printed. Aqueous or NMP/water MXene inks for electrodes are established [Zhang et al.(2019), Spurling et al.(2021)]. Shear during coating can align MXene *in the electrodes*, where electronic percolation is desired. Alignment through the separator thickness is not desired.

Drying: staged ramp to 80–120 $^{\circ}\text{C}$ under inert or dry gas. Optional electrode anneal ≤ 250 $^{\circ}\text{C}$ under argon if the binder survives; the electrolyte membrane is not subjected to that step after

LiTFSI is introduced. Lamination: 0.5–1 MPa at 70–90 °C. Assembly in a dry room. Formation at low rate (C/20–C/10) with voltage cutoffs set from three-electrode or half-cell windows.

Graded electrode porosity (more open near the separator) is optional and standard porous-electrode practice. It is not assigned a unique theoretical origin.

5 Conservative performance envelope

Table 1 is an order-of-magnitude bill of materials for a single-layer printed cell of 150–180 μm total thickness, excluding pouch foil. Densities are approximate packed densities after drying.

Table 1: Illustrative areal bill of materials for a prototype printed cell.

Layer	Thickness (μm)	Areal mass (mg cm^{-2})	Role
Al foil (pos.)	8	2.2	collector
Positive electrode	50	6.0	V_2O_5 hybrid
Separator	30	4.0	PVDF–PEO–LLZO
Negative electrode	40	4.0	$\text{Ti}_3\text{C}_2/\text{Nb}_2\text{C}$
Al or C collector (neg.)	8	2.2	collector
Stack total	136	18.4	—

If the positive electrode delivers 200 mAh g^{-1} of hybrid and the hybrid is 70 wt% of the 6.0 mg cm^{-2} layer, areal capacity is $\sim 0.84 \text{ mAh cm}^{-2}$ before matching. The negative electrode at 180 mAh g^{-1} of 80 wt% active in 4.0 mg cm^{-2} is $\sim 0.58 \text{ mAh cm}^{-2}$. The cell is then negative-limited at $\sim 0.58 \text{ mAh cm}^{-2}$. At 2.1 V average, areal energy is $\sim 1.2 \text{ mWh cm}^{-2}$. Divided by 18.4 mg cm^{-2} this is $\sim 65 \text{ Wh kg}^{-1}$ at the stack level before packing. Raising negative loading, lowering porosity, thinning the separator to 20 μm , and achieving 2.3 V average can push the same chemistry toward $\sim 100\text{--}150 \text{ Wh kg}^{-1}$. Approaching 180 Wh kg^{-1} requires optimistic capacities and very little inactive mass. Values above 250 Wh kg^{-1} require a different positive electrode (high-Ni oxide or sulfur) and usually a lithium-metal or lithium-alloy negative electrode. They are outside this materials set.

Rate: MXene negatives are fast in liquid electrolyte. A 30 μm polymer separator at $10^{-4} \text{ S cm}^{-1}$ has an area-specific resistance of order 30 $\Omega \text{ cm}^2$. At 1 mA cm^{-2} ($\sim 2\text{C}$ on a 0.5 mAh cm^{-2} cell) ohmic drop is tens of millivolts; at 5 mA cm^{-2} it is already large. High-C operation is therefore plausible only at low areal loading. A design target of 1–3C continuous for a first solid printed cell is honest. 10C-class charging is a later measurement, not a specification.

Cycle life: [Myint et al.(2024)] observed substantial fade of the V_2O_5 hybrid at 1 A g^{-1} . MXene negatives can cycle thousands of times in a capacitive window. The full cell will be limited by the positive electrode and by the solid interfaces. A first-generation target of hundreds to a low thousands of cycles at shallow depth of discharge is consistent with the cited hybrids. Claims of 10^5 cycles are not.

Operating temperature: cross-linked PVDF–PEO with 10 wt% LLZO may function near room temperature. Operation at -45°C without a plasticizer is not assumed. An upper limit near 80–90 $^\circ\text{C}$ is set by PEO and by MXene oxidation risk, not by 110 $^\circ\text{C}$.

6 Measurements that would support or falsify the design

1. Electronic conductivity of the separator by DC polarization. If σ_e is not at least three orders of magnitude below σ_{ion} , the MXene loading is too high.
2. Half-cell voltage windows and reversible capacities of each printed electrode versus Li metal in liquid electrolyte, then versus the solid membrane.
3. Full-cell open-circuit voltage and first-cycle loss after a stated prelithiation protocol.
4. Stack-level Wh kg⁻¹ computed from measured capacity, average voltage, and weighed layers—not from active-mass-only half-cell numbers.
5. Post-mortem inspection for MXene oxidation, V₂O₅ dissolution or amorphization, and interfacial voids after cycling.

Failure of (1) or (3) ends the design. Failure of (4) to exceed ~ 80 Wh kg⁻¹ does not; it merely sets the application (printed hybrid capacitor / moderate-energy solid cell) rather than a high-energy traction cell.

7 What this chemistry is for

The honest application is a fully or mostly printed, lithium-ion or lithium-ion-capacitor cell that uses abundant transition-metal carbides and a vanadium oxide hybrid, accepts a modest voltage, and trades energy density for printability and rate in thin layers. That is a real materials niche. It is not a substitute for NMC/graphite, LFP, or lithium-metal solid-state cells on energy density.

8 Conclusion

From the printed MXene hybrid recipes under discussion, the combination that survives electrochemical constraints is an asymmetric lithium-ion cell: a Ti₃C₂T_x/Nb₂CT_x negative electrode, a V₂O₅/TiO₂@MXene positive electrode, a PVDF-PEO-LiTFSI-LLZO separator containing only a few percent MXene, and an explicit lithium inventory. Process temperatures must remain inside the polymer window. Literature capacities support a functioning laboratory cell. They do not support 400–650 Wh kg⁻¹, 10–17C continuous charging of a high-energy stack, or 10⁵-cycle life. Those figures should not be attached to this bill of materials. The next step is half-cells and a capacity-matched full cell, not a further increase in projected metrics.

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